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How much does volatility influence stock market returns? Empirical evidence from India

Malvika Saraf, Parthajit Kayal*

Madras School of Economics (MSE), Kottur, Chennai, Tamil Nadu, India

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Abstract The purpose of this paper is to establish and estimate the extent of the volatility anomaly (VA). We examine the impact of the beta, variance, relative-beta, and relative-variance measures on the stock returns for NIFTY500 companies, for the 10-year period 2010-2020. Our empirical findings suggest that the VA is predominant in the medium to long-term, but seems to be negligible in the ultra-short and short time frames. The overall findings suggest that the VA is most significant when the time period considered is three years or more. These results can be highly useful for investors as well as portfolio managers.
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Introduction

India is among the fastest-growing economies in the world, with a colossal amount of foreign direct and indirect investment in the country. Hence, it is important to examine whether these resources are being allocated efficiently and if this is reflected in stock prices. The growth of the Indian stock market has been more than impressive in the last two decades, especially in terms of market capitalisation, number of listed firms, and turnover rates. Numerous studies have investigated the efficiency of the Indian stock market (Dicle, Beyhan, & Yao, 2010, Kumar, Mamidi, & Marisetty, 2011; Narayan & Ahmed, 2014; Narayan, Ahmed, & Narayan, 2015; Narayan, Narayan, & Prabheesh, 2014). Evaluating stock market dynamics in the Indian context deserves attention for several reasons. First, as a developing nation, India has become the economic powerhouse of the world and has endless potential to expand and deliver more. Second, the National Stock Exchange (NSE) is the largest derivatives exchange in the world in terms of the volume of contracts traded, the second largest exchange in the world in terms of the number of contracts traded in single stock futures, and ranked first in terms of contracts traded in index options and currency futures. Third, according to the International Monetary Fund, India is the 11th largest economy in the world by nominal gross domestic product and the third largest by purchasing power parity. Over the past two decades, the stupendous growth of the Indian stock market has been accompanied by a surge in low-risk/high-return investment strategies as well as in the volume and size of investments by both domestic and international investors. Hence, studying its volatility anomaly (VA) and thereby the risk-adjusted expected return pattern is very important to support investors in making their investment decisions and monetary policy specialists for policy formulation. Furthermore, the study would be relevant for investors who may wish to invest in Asian countries. As most Asian countries (except a few)

*Corresponding author: Tel. No. (91) 044 - 22300304
E-mail address: parthajit@mse.ac.in (P. Kayal).

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are emerging in nature, the VA is not a phenomenon confined to just India. We would expect similar return patterns across many other countries in Asia. Foreign and domestic investors can use this anomaly to develop low-risk-high-return (LRHR) strategies to invest in Asian financial markets.

The relationship between risks and returns lies at the heart of modern finance. Traditional finance gurus often talk about the widely prevalent notion that high risks could yield higher returns. In other words, stocks exhibiting higher volatility measured by variance and stocks with higher systematic risk measured by beta could yield higher returns for investors. But evidence from equity markets all over the world has presented quite a different picture in the last two decades. The capital asset pricing model (CAPM), originally put forward by Lintner (1965) and Sharpe (1964), postulates that the risk of an asset should be measured by the extent to which it contributes to the volatility of a well-diversified portfolio. According to the CAPM, higher beta assets should demand a higher average return to compensate investors for bearing higher risk. In contrast, the actual relationship between beta and return is much flatter than predicted by the CAPM (Black, 1972; Jensen et al., 1972; Haugen & Heins, 1975). Motivated by empirical results, Baker, Bradley, and Wurgler (2011) contended that high-beta and high-volatility stocks consistently underperformed compared to low-beta and low-volatility stocks. Their findings not only contradicted CAPM but also clearly underscore the very proven principle behind risks and returns: low-risk portfolios seem to have yielded greater rewards than high-risk portfolios.

The observation that low-beta stocks outperform high-beta stocks has received a lot of research attention and triggered the need for a profound investigation into the same, especially in the context of the Indian stock market. More specifically, low-volatility stocks, i.e., stocks which fluctuate less over time and have lower variance and lower beta, are the ones that have provided a higher expected return to investors in the long run. This long-term risk-return relationship is termed the VA (De Jong & Palkar, 2016). Substantial evidence from international markets like the United States and China shows that future stock returns of previously low-return variability stocks significantly outperform those of previously high-return variability stocks. Consistent with this view, this paper showcases how the average returns on stocks within a specified period can be explained in the context of the Indian stock market using symmetric and asymmetric risk measures such as (1) variance, (2) beta, (3) relative variance, (4) relative beta, (5) downside beta, (6) downside semi-variance and (7) value at risk. Our main goal throughout the paper is to understand the implication of the first four risk measures on stock returns; the other two are considered for cross-verifying our results.

Our theoretical analysis begins by studying the dynamics of risk and why the aforementioned risk standards are chosen to explain the trend pattern in expected stock returns. Risk pertains to the uncertainty of getting back the money (or even more) one invests. It should not be surprising that most investors are more concerned about downside risk than upside risk. The downside risk is the risk of the actual return being below the expected return. It is common knowledge that India mostly comprises institutional investors who are risk-averse. They certainly want to make a profit, but at the same time, they do not want to incur losses. According to Estrada (2002, 2007) and Post, Van Vliet, and Lansdorp (2012), downside risk measures fare better than traditional risk measures in explicating the variability in long-term stock returns.

We shed light on two categories of risk measures. The symmetric and standard risk measures include variance and beta. Asymmetric risk measures such as relative variance and relative beta are used to account for downside risk. Ang, Chen, and Xing (2006a) delineated three proxies for downside risk: downside beta, net beta (upside beta minus downside beta), and relative downside beta. While downside beta is the most appropriate measure of downside risk, future stock returns cannot be sorted on this basis because, by construction, regular beta and downside beta are not independent of each other. Thus, relative variance (difference between negative semi-variance and variance) and relative downside beta (difference between downside beta and beta) qualify as the most appropriate measures of downside risk since they accurately capture the incremental impact of downside beta over regular beta and downside variance over regular variance.

We establish our empirical results for a cross-section of stocks listed on the NSE (NIFTY500 index) over the 10-year period from July 2010 to June 2020. From the daily stock price data of these companies, we calculate the daily stock returns. Going forward, we will use various risk measures to examine the pattern of mean returns on these stocks. For each risk proxy, we divide the data into five quintiles, from '1' being the lowest to '5' being the highest. Subsequently, for each risk measure, we check whether low-quintile stocks provide higher mean returns than high-quintile stocks and whether the VA manifests. Furthermore, we comprehensively examined the 10-year data through several robustness checks by undertaking the same procedure on a rolling-year basis for different subsamples.

Our approach to checking for higher returns yielded by stocks bearing higher volatility and systematic risk closely follows Ang, Hodrick, Xing, and Zhang (2006b). We have differentiated our work from its companion papers by conducting our study on a rolling year basis for different time horizons, namely, very short term (less than a calendar year), short term (more than 1 year and less than 3 years), medium term (more than 3 years and less than 5 years), and long term (5 years and more). Finally, we also examine the data from the 6-month COVID pandemic phase from January to June 2020. Our findings suggest that the VA is more of a medium- to long-term phenomenon than a short-term one. A possible explanation is that the market typically requires 3 years or more for risk adjustment and volatility smoothing to yield higher returns.

Modern investors are perpetually concerned with risk-adjusted returns. The low-risk anomaly explanations focus on an investor's risk appetite. Time-varying market volatility alters their investment opportunity set by influencing the expectation of future market returns or changing the risk-return trade-off. Hence, the importance of understanding stock returns in terms of risk measures cannot be stressed more. The bonus from the LRHR strategies is attributable to the compounding effect – the lower volatility pressure on investment returns enhances the performance of less

volatile stocks. The real benefits of possessing low-volatility stocks accrue over longer periods of time, as the well-known power of compounding suggests (Ang et al., 2006).

This work proceeds as follows. We present the literature review in the following section. We elaborate on the data and methodology in the third section, where we discuss the source of data, selection, and construction of variables. Subsequently, in the fourth section, we present our empirical results. In the fifth section, we conduct extensive robustness checks as well as pinpoint potential extensions. The sixth section concludes the paper.

Literature review

An enormous amount of research has been devoted to the fact that around the world, low-volatility stocks have higher average returns than high-volatility stocks (Ang et al., 2006b; Ang, Hodrick, Xing, & Zhang, 2009; Baker & Haugen, 2012; Baker et al., 2011; Blitz & van Vliet, 2007; Clarke et al., 2006). The outperformance of low-volatility stocks over high-volatility ones is economically exceptionally large, amounting to 12% per annum on average (Walkshäusl, 2013). As a result, Baker et al. (2011) proclaimed that 'the outperformance of low-risk portfolios is perhaps the greatest anomaly in finance'. Baker et al. (2011) as well as Frazzini and Pedersen (2014) found compelling evidence of this anomaly in the US stock market data. Furthermore, Collver, De Jong, and Palkar (2013) as well as Baker and Haugen (2012) observed that results from global equity markets mostly concurred with those of the United States when they tested the VA in multiple developed and emerging countries. While investigating the same in Germany's stock market, Koch (2010) discovered strong evidence that low-risk stocks generate considerably higher returns than high-risk ones by around 9% per annum. Consistent with the preceding evidence, Saengchote (2017) detected that Thai stocks with low beta exhibited positive abnormal returns, whereas high beta stocks exhibited negative abnormal returns for the 12-year period of 2004–2015.

There are multiple reasons underlying the existence of the VA. For example, Baker et al. (2011) argued that the low volatility effect arises because sophisticated investors must follow a benchmark and, consequently, are unable to take advantage of an arbitrage opportunity, whereas it is possible to systematically earn higher returns while assuming lower risk. This anomaly can persist because institutional investors cannot fully exploit the excess returns they could gain from investing in such stocks. 'Limits on arbitrage' (Dutt & Humphery-Jenner, 2013) could be another reason for this anomaly. However, they pointed out that 'limits on arbitrage' would be less dominant in emerging markets as an explanation for the VA. This is mainly due to the existence of fewer and more stable stocks in the emerging market equity index and the limited availability of quality stocks for investments. It encourages large investors to choose low-volatility, stable stocks and thereby arbitrage any potential excess profits. Furthermore, media attention can also have an impact on investors' attention and influence their choices. In this case, also, the VA disappears for stocks with high media attention (Blitz, Huisman, Swinkels, & van Vliet, 2020) as too much attention could arbitrage away potential excess profits. However, it is also true that, in the case of arbitrage asymmetry, limited arbitrage opportunity would eliminate more underpricing than overpricing (Stambaugh, Yu, & Yuan, 2015). Therefore, it is generally observed that quality stocks in emerging markets are overpriced, stable, and create high risk-adjusted returns. That possibly explains the reason for the VA in emerging markets like India. Similarly, different forms of regulations and constraints on short selling (which is prevalent in most of the emerging stock markets, including India) support the argument of arbitrage asymmetry that leads to the VA (Stambaugh et al., 2015). Also, trading frictions in emerging markets could also be high, and therefore arbitrage can be costly (Shleifer & Vishny, 1997). This will again possibly contribute in favour of the VA argument. Another reason this anomaly exists could also be irrational investors' sentiment and its impact on volatility. In a recent study, Haritha and Rishad (2020) highlighted that positive sentiment due to irrationality could lead to higher speculative activities and thereby overvaluation of stocks. This sentiment also reduces volatility in the long run due to the bullish outlook on overvalued stocks. Although the authors' argument wasn't directly motivated to establish the VA, we could identify a clear connection between volatility and returns.

Risk-based explanations pose certain difficulties in describing the observed return pattern, as the return difference between low- and high-risk stocks cannot be accurately captured by common asset pricing models. This is precise because low-volatility stocks typically have low market betas, whereas high-volatility stocks exhibit high market betas. Consistent with this view, Black et al. (1972) were among the first to contend that high-beta stocks have lower returns than equilibrium and that low-beta stocks earned higher returns than expected. This gives rise to the fact that low-risk stocks should be considered a unique asset class while strategically allocating assets (Blitz & van Vliet, 2007).

This study is strongly connected with that of Ang et al. (2006b). They acknowledged that stocks that are highly sensitive to innovations in aggregate stock market volatility have low mean returns. They further point out that using other measures of aggregate volatility risk (such as sample volatility, extreme value volatility estimates, and realised volatility estimates calculated from high-frequency data) results in little spread in the cross-section of average stock returns. Our main upshot is that both beta and variance are significantly valued risk factors in the cross-section of returns.

Our paper concentrates on the extensive impact of low volatility and low risk on expected stock returns within a time frame. The appealing feature of our medium of analysis is to study this impact with respect to risk-based measures. In congruence with this, Baker et al. (2011) posited behavioural explanations for this anomaly. They contended that the volatility effect may be partially elaborated by the irrational preference for high-volatility stocks by individual investors and partially by the institutional investor's mandate to beat a given benchmark which tends to limit investments in low-volatility stocks. In addition to this, the 'low volatility effect' seems to be related to operating performance and investment. Less volatile stocks would most likely exhibit strong operating performance as low volatility

improves a firm's access to capital. In an efficient market, there should be a connection between stock returns and (positive) earnings surprises, but not merely between stock returns and earnings per se (Core et al., 2006).

Finally, our study is also related to the literature on rare and unexpected events. Starting with Rietz (1988), many researchers have modelled the possibility of the occurrence of rare disasters, such as wars, global pandemics, or economic depressions, to unfold the equity premium puzzle (e.g., Barro, 2006; Gabaix, 2008, 2012). Similarly, this paper also investigates the impact of the coronavirus pandemic on stock returns. A point worth noting is that such unexpected events or disasters are extremely rare.

Data and methodology

Data

The data in this study consists of daily stock prices for 362 companies partly constituting the NIFTY500 index for the 10-year period starting from 1 July 2010 to 30 June 2020. Daily stock price data is not available for many companies prior to 2010. This sample represents approximately 72% of the overall population, which is a large proportion to work with. Increasing the time frame would further reduce the sample size. The period under study is chosen with the objective that stock price fluctuations can be observed for certain ups and downs in the market, such as the post-2008 financial crisis period and the period of the coronavirus pandemic starting from December 2019. Also, a 10-year period would possibly reflect both bear and bull market scenarios.

Methodology

The standard risk measures used in this study include variance and beta. Variance is calculated for each company's stock price by taking the average of squared deviations from their mean returns. Variance explains the degree of spread or variability in the data. Beta is a measure of the systematic risk of a stock compared to the market as a whole. It is calculated as the ratio between the covariance of expected returns on a stock and returns on the market, and the variance of average expected returns on the market.

$$\beta_j = \frac{Cov(market,\ stock)}{Var(market)} = \frac{cov(\sim r_j, \sim r_M)}{\sigma_M^2} \tag{1}$$

where β = beta; Cov(market,stock)= covariance between market and stock and Var(market) = variance of market.

Following this, to evaluate the effectiveness of relative variance and relative beta on mean returns, we should calculate the downside deviation, i.e., the variance of the negative returns to the stocks. Relative variance is defined as the difference between variance and downside variance (or deviation). Downside beta is one of the most direct measures of downside risk. It is calculated as the ratio of the covariance of the expected return on a stock and the return on the market and the downside deviation. Relative beta is defined as the difference between downside beta and beta. It underestimates the risk for low-beta stocks and overestimates the risk for high-beta stocks. This explains why low-beta stocks seem to be systematically underpriced and high-*beta* stocks seem overpriced in empirical tests of CAPM in the mean-variance framework (Black et al., 1972; Fama & French, 1992; Fama & MacBeth, 1973; Reinganum, 1981).

The observation that investors are more worried about downside risk than upside risk when it comes to decision-making leads to the possibility that they might treat downside risk differently from market risk. In this context, Markowitz (1959) suggested that downside semi-variance would be a more appropriate risk measure as compared to variance. Downside semi-variance is defined as the variability of a stock up to its mean level. Compared to variance, downside semi-variance concentrates on the bad outcomes of stock returns. Mathematically, it is represented as follows:

$$SV(<) = \frac{1}{n} \sum_{j=1}^{n} \left(r_j - r_j\right)^2 \tag{2}$$

where each observation r_j *i*s lower than the mean vale of all observations (r_j).

Another integral measure which is related to the risk of loss of investments is the value at risk (*VaR*). This *VaR* is defined as the maximum amount of loss that one can expect for a given confidence interval and a given time horizon. Suppose a portfolio has a 4.3% *VaR* at a 5% probability level for 1 month. This would imply that over the next month, there is a 5% chance that the portfolio could lose more than 4.3% of its value, or there is a 95% chance that the portfolio will lose less than 4.3%. Also, a positive *VaR* indicates a loss, whereas a negative *VaR* indicates a profit. Our calculation of *VaR* at a 95% confidence limit for the data under study shows negative values, indicating a profit for these stocks. Mathematically, it is represented in Eq. (3).

$$VaR_\alpha^{\%} = z_\alpha \sigma \tag{3}$$

We begin our analysis by investigating the risk-return relation for the constituent stocks of the NIFTY500 Index. To explain the low volatility–high return relationship, we group the stock return data in terms of the highest mean returns and then sort them based on each of the risk measures – variance, beta, relative variance, and relative beta. Furthermore, we divide all stocks into five quintiles, from 1 being the lowest quintile to 5 being the highest. Based on the quintile breakpoints for the variables, stocks are segregated into low- and high-volatility stocks, as well as safe and risky stocks. We find that low beta and low-volatility stocks (and not the lowest beta and least volatile stocks) generate high expected returns within the mentioned time frame. To provide more clarity, stocks falling under the first three quintiles are considered low-beta stocks since their beta values are roughly below the market beta and the ones falling under the fourth and fifth quintiles are high-beta stocks with beta values usually above the market beta. A similar description holds true in the case of low- and high-volatility stocks.

In addition to the 10-year period of analysis, we cross-verify the presence of the VA by conducting the same procedure over a period of 7 years, 5 years, and 3 years on a

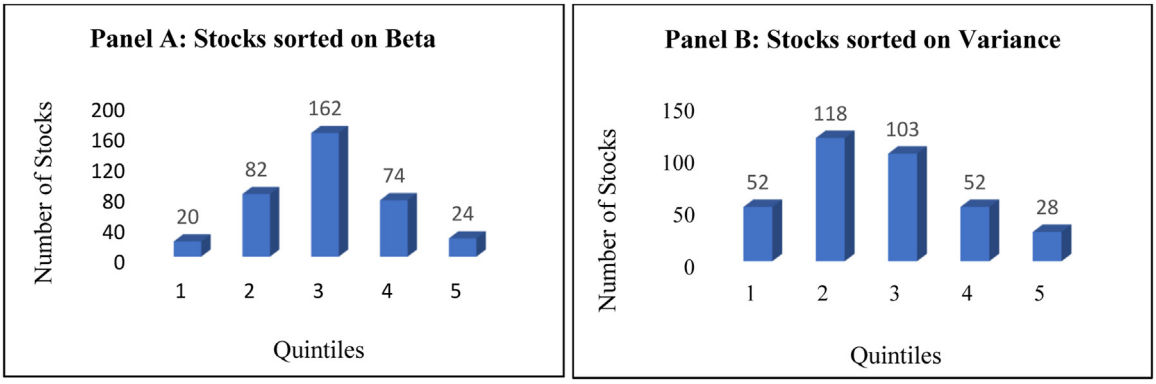


Figure 1 Stocks of companies sorted on beta and variance. Panel A and Panel B exhibit the number of stocks in each of the five quintiles following sorting by beta and variance, respectively.

rolling basis as well as for a period of 6 months when the world was hit by the unexpected Coronavirus pandemic.

Empirical results

Our theoretical reasoning coupled with a study of the past literature clearly suggested that low-beta stocks and low-variance stocks outperform the high-beta and high-variance ones. In this section, we provide strong empirical support for the VA. Initially, we sort the stocks based on the beta, variance, relative beta, and relative variance and present the existence of the VA in each case. Later, we sort the stocks based on *VaR*, downside semi-variance, and downside beta and provide interpretations.

Stocks sorted by beta and variance

To check the existence of VA, we sorted the stocks of 362 companies of beta and variance. Early studies that explain the empirical failure of the CAPM include Brennan (1971), Black (1972), Jensen et al. 1972, and Haugen & Heins (1975). Later, Ang et al. (2006, 2009) argued that volatility gives negative premiums for equity returns and that stocks with high sensitivity to aggregate volatility risk earn low returns. Our findings help elucidate these anomalous relations between volatility and stock returns.

Figure 1 presents the graphical pattern of sorting stocks on the basis of beta and variance. Panel A shows the number of stocks in each quintile when they are beta-sorted. Out of 362 companies, the stocks of 263 companies that form the low-beta quintiles (first, second and third) give higher mean returns as compared to stocks in the high-beta quintiles (fourth and fifth), as highlighted in Figures 1 and 2. The VA holds true for this risk proxy since a low beta, i.e., low systematic risk, is associated with higher expected returns in the future (Baker & Haugen, 2012). Panel B shows the number of stocks in each of the five quintiles when they are sorted according to variance. While dividing all the variance values into five quintiles, we find that all the stocks are confined to three quintiles (first, second and third). Out of 362 stocks, the lowest quintile giving the highest mean returns contained 353 stocks, which makes it difficult to make any inferences. To resolve this issue, we further divided these 353 stocks into five quintiles. Out of these, stocks of 273 companies lie in the low variance quintiles (first, second and third) and display higher mean returns as compared to stocks falling under the high-variance quintiles (fourth and fifth), as displayed in Figures 1 and 2. The VA exists since low variance, i.e., low risk and less stock price fluctuation, is associated with high average returns. In the case of both these risk

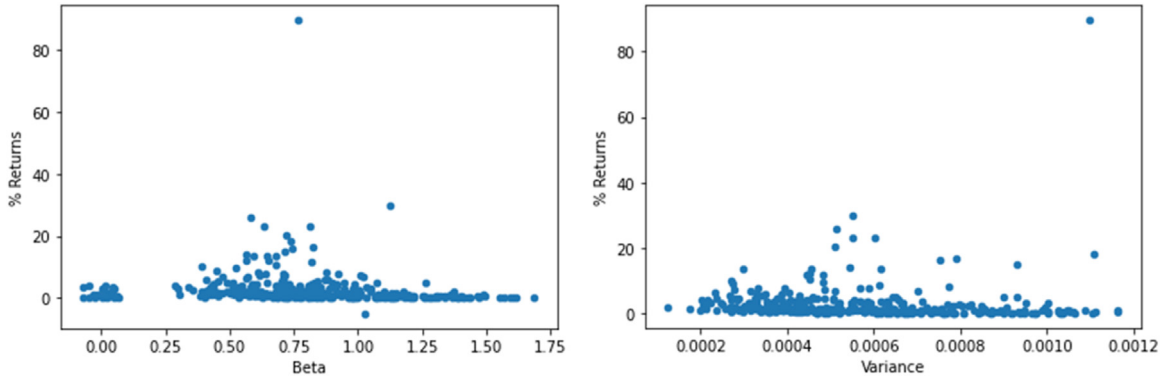


Figure 2 The pattern of returns (in 100s) for all beta and variance values. The left and right boxes present the return pattern for all beta and variance values, respectively. The returns for beta mostly lie within the range of 100%–2500% (second and third quintiles). The right box clearly shows that low-volatility stocks provide high average returns. Thus, the mean returns are higher for lower beta and variance values, validating the presence of VA.

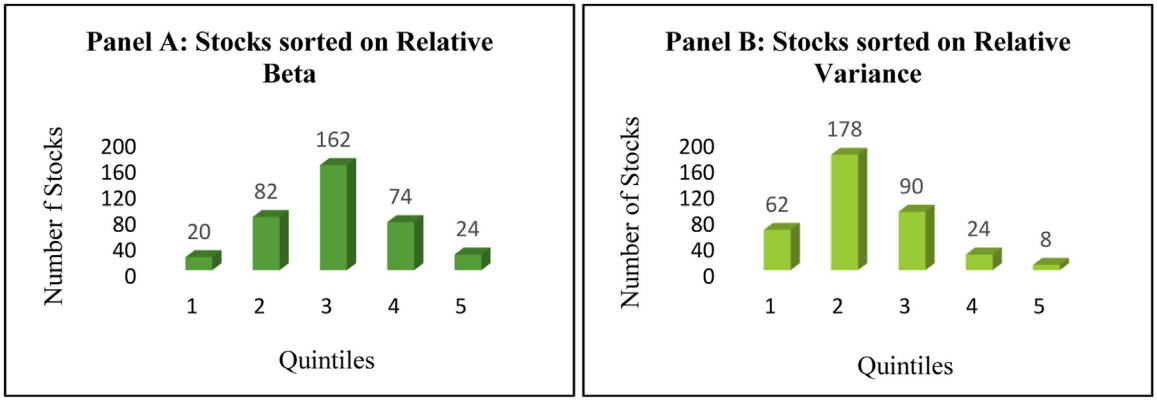


Figure 3 Stocks of companies sorted on relative beta and relative variance. Panel A and Panel B exhibit the number of stocks in each of the five quintiles following sorting by relative beta and relative variance, respectively.

measures, our empirical findings are consistent with the argument that the low-beta and low-variance stocks outperform and provide higher mean returns in the long run as compared to the high-beta and high-variance stocks (Baker et al., 2011; De Jong & Palkar, 2016; Heston, Rouwenhorst, & Wessels, 1999; Lakonishok & Shapiro, 1984).

Stocks sorted by relative beta and relative variance

While dealing with the cross-sectional data, the importance of downside risk measures cannot be stressed. Hence, we need risk measures that specifically take into consideration the liability traits of investment decisions (Leibowitz, 1986). In this section, we re-examine the risk-return relation, using asymmetric risk proxies like relative beta and relative variance. By construction, quintile 1 represents the stocks with the lowest downside risk, making them the least risky, and quintile 5, on the other hand, represents the stocks with the highest downside risk, and thus the riskier ones.

Figure 3 shows the results of sorting the stocks based on relative beta and relative variance. Panel A shows the number of stocks in each quintile when they are relatively beta-sorted. We observe the exact same return pattern as for the beta-sorted stocks, as displayed in Figures 1 and 2. The VA is predominant here since low-relative beta signifies lower downside risk and shortfall which is associated with a higher expected return here. Panel B shows the number of stocks in each of the quintiles when they are sorted by relative variance. Out of the 362 companies, the stocks of 241 companies belonging to the low-relative variance quintiles (first, second and third) compound into higher mean returns over time as compared to stocks under the high-relative variance quintiles (fourth and fifth). This is graphically represented in Figures 3 and 4. The VA evidently holds true in this setting since low-relative variance signifies more safety which is associated here with a higher expected return (Grootveld & Hallerbach, 1999). We can observe that the lowest-quintile stocks are safer than the highest-quintile ones in terms of future expected returns. High-risk stocks exhibit much higher time-series variation, resulting in lower compound returns over time (Bollerslev, 1987; Vuolteenaho, 2002). In the backdrop of both these risk measures, we can comfortably say that the low relative beta and low relative variance stocks are safer as well as fare better than the high relative beta and high variance ones in terms of giving higher expected returns (similar to the findings by Fong, 2018; Shefrin, 2001).

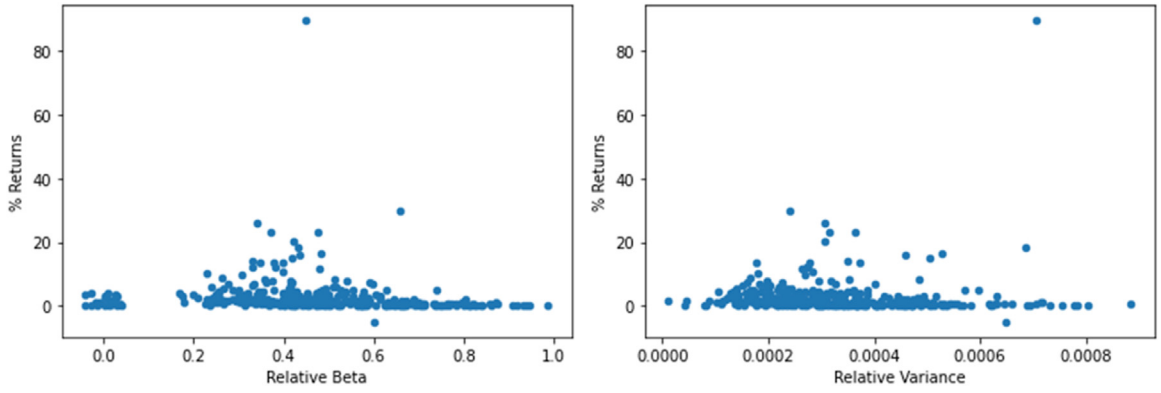


Figure 4 The pattern of returns (in 100s) for all relative beta and relative variance values. The left and right boxes present the return pattern for all relative beta and relative variance values, respectively. Returns mostly lie within the range 100%–2500% for relative beta values in the range 0.25–1.00 (2nd and 3rd quintile). The right box unveils the return pattern for all relative variance values. In the right box, returns mostly fall within the range of 100–3000%. We can easily comment that less volatile stocks provide higher mean returns, validating the prominence of the VA.

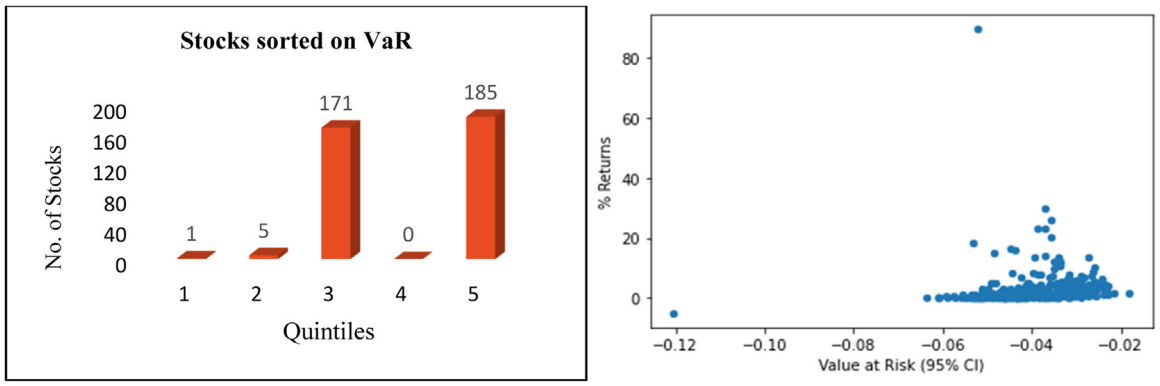


Figure 5 The pattern of returns (in 100s) for all stocks sorted on *VaR* values. The left box exhibits the number of stocks in each quintile when sorted by *VaR*. Since we obtain negative values from *VaR* calculations, the higher the quintile or value of *VaR*, the higher the expected future return. We find that 355 stocks giving higher mean returns lie within the higher *VaR* quintiles (third and fifth). The box on the right shows the return pattern for all *VaR* values. We observe that the highest expected returns are obtained for stocks belonging to the third and fifth quintiles. *VaR*, value at risk.

Stocks sorted by *VaR*, downside semi-variance and downside beta

A related analysis can be made for downside risk measures like *VaR*, downside semi-variance, and downside beta. *VaR* shows the worst expected loss over a target horizon within a given confidence interval. It summarises the exposure to market risks and the probability of adverse changes in financial variables in a single number. Semi-variance (Markowitz, 1959) measures the variability of returns below the mean. Downside beta is the sensitivity of stock returns to the market. It may change with the level of the market returns, leading to an asymmetric risk profile (Post et al., 2012).

VaR is the maximum loss that a firm can observe within a specified confidence limit and for a given time horizon. Here, we consider the confidence limit to be 95%. *VaR* estimation highly depends on a good forecast in the light of unexpected and rare events, or catastrophic risks, since it is quantified from the lowest stock returns (Danielsson & De Vries, 2000). A positive *VaR* indicates a loss, whereas a negative *VaR* indicates a profit. Since our *VaR* calculations generate negative values, the higher the *VaR*, the higher the profit and also, therefore, the expected returns on the stocks. This concept is projected in Figure 5. Out of the 362 stocks, 355 belong to the third and fifth quintiles. The pattern of mean returns is expressed in the right box of Figure 5.

Downside semi-variance measures the dispersion of returns below the expected value of an investment return. Measuring downside risk using semi-variance provides an intuitive perception of the risk borne by investors (Boasson, Boasson, & Zhou, 2011). We know that semi-variance is calculated below a targeted level of return. Here, the benchmark level of return is taken as the mean of the variances of all stocks, so the values represent returns below the mean of variance. The resultant values are plotted graphically as shown in the right box of Figure 6. In contrast with the results of all the previously used variables, the VA is not observed. Out of 362 stocks, 315 lie in the fifth quintile, which exhibits neither the highest mean returns nor the least standard deviation of returns, as shown in the left box of Figure 6. Additionally, the graph on the right box of Figure 6, showing the return pattern for all values of semi-variance, clearly points to the fact that higher risks provide

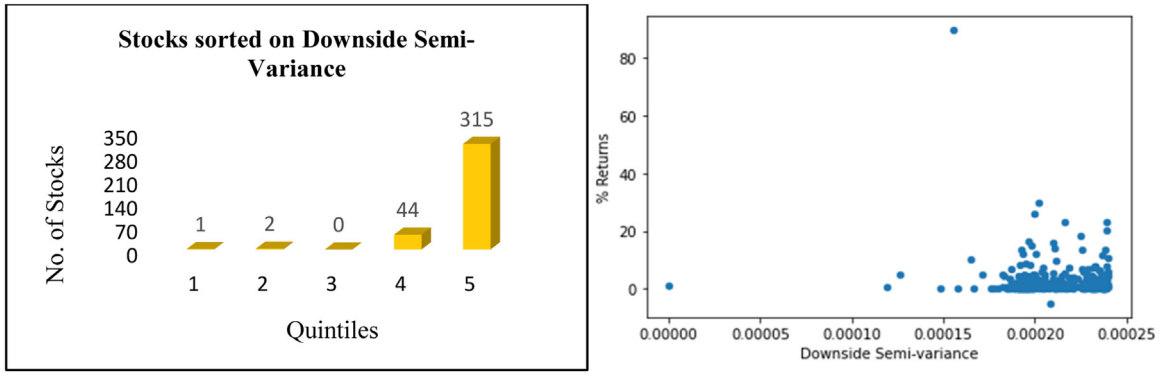


Figure 6 The pattern of returns (in 100s) for all stocks sorted on downside semi-variance values. The left box of this figure shows the number of stocks in each quintile when sorted by downside semi-variance. The right box shows the return pattern for all downside semi-variance values. Since the returns are observed to be higher for higher values of semi-variance, we find that the VA is non-existent for this risk measure over the 10-year sample period. These results are consistent with the traditional notion of an association between higher risks and higher returns.

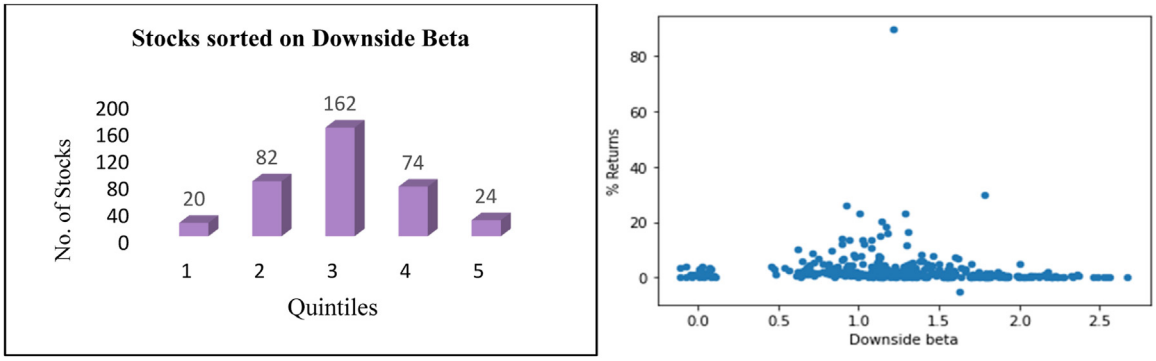


Figure 7 The pattern of returns (in 100s) for all stocks sorted on downside beta values. The left box in the figure exhibits the number of stocks in each quintile when they are sorted by downside beta. Ignoring the extreme value, the dispersion of returns is within the range of 100–2500% for downside beta values in the range of 0.50–1.75 (second and third quintile). The right box shows the return spread for all downside beta values. We distinctly observe that the highest expected returns are obtained for stocks belonging to the second and third quintiles, validating the VA claim.

higher returns over the 10-year sample period considered. Contrary to our expectations, the data provide evidence against the VA for this downside risk measure.

Downside beta is defined as the susceptibility of excess stock returns to excess market returns during market downturns. A market downturn is a period during which the excess market return is lower than its mean value during the past year (Atilgan, Bali, Demirtas, & Gunaydin, 2018). The stocks of the 362 companies are sorted into five quintiles based on the values of downside beta. We can detect that these results are exactly the same as in the case of the beta and relative beta. This is not surprising since relative beta is the difference between downside beta and beta. This is clearly projected in Figure 7, along with the overall spread of returns for all values of downside beta. Our data provide persuasive evidence in favour of the VA for this risk measure.

From these results, it is quite evident that the presence of VA is strong in the Indian stock market. A plausible reason is the limited availability of quality, stable, and liquid stocks. The Indian stock market has approximately 6000 stocks, out of which only 10%–15% are purely liquid stocks. Out of these, only around 100 stocks are of high quality (Walkshäusl, 2013). Hence, retail and institutional investors prefer to invest only in these stocks. They are safe stocks in the sense that they are less volatile and yield decent and consistent returns over a long period of time. The equity market is divided into large, mid and small-cap stocks. Most of the safe stocks belong to the large-cap category. Since investors mostly invest in liquid stocks giving satisfactory returns during their investment periods, occasionally we witness excess demand in the market on the one hand and no demand for these stocks on the other. This is the primary reason why volatility fluctuates so much in the Indian stock market. When we invest in such stocks over a long period, we find that the volatility is smoothed out and high average returns are earned by bearing lower idiosyncratic risk.

Furthermore, we also observe some firm-level characteristics that are related to the VA. Since we focused on the documentation of the VA in India using different methods, we have not specifically analysed the firm-level characteristics in our study. However, carefully observing the set of companies for which we consistently observe high returns and low volatility, we find that most of the companies are high quality in nature (irrespective of small or large) indicated by their high profitability ratios, stable free cash flows, high returns on capital employed, low debt-to-equity ratios, high level of promoters' holdings, and such indicators. This observation is very similar to the findings of Walkshäusl (2013) as he observed the VA for high-quality firms.

Robustness check

From our analysis using a cross-section of companies from the NIFTY500 stock index for a period of 10 years, we learned that the VA clearly exists in the Indian stock market. As Warren Buffet observed, 'Risk cannot be defined until it has a specified time horizon'. This makes it extremely important to check whether the VA persists over different time frames and to varying extents. Here, we consider sample periods of 7 years, 5 years, and 3 years on a rolling basis and the 6-month COVID pandemic phase from January to June 2020. The choice of 7-, 5-, and 3-year rolling windows is to show how VA performs over different time horizons (medium-term to long-term). The same process can also be repeated with 2, 4, 6, and 8 years. We have cross-checked the results and found them to be similar, and VA holds for 4-, 6-, and 8-year windows.

Expected stock returns have been observed to change with time so that the investment opportunities are not constant (Campbell & Viceira, 1999). This is due to the stochastic nature of expected returns, which enables us to understand their behaviour over time. Table 1 shows the number of stocks in each quintile sorted on beta, variance, relative beta, and relative variance for 7 years on a rolling basis from July 2010 to June 2020. There are four different sample data sets generated in this case, i.e., 2010–2017, 2011–2018, 2012–2019, and 2013–2020. Figure 8 exhibits the distribution of stocks giving the highest mean returns into quintiles when they are sorted on beta values. Figure 9 displays the return pattern of stocks for all values of beta, variance, relative beta, and relative variance for the 7-year period of 2013–2020.

Table 1 Number of stocks yielding the highest mean return sorted on various risk measures (7 years).

Highest mean return						
	Low				High	
Quintiles 7-year Rolling	1	2	3	4	5	Low > High
Panel A: Beta sorted						
2010-2017	35	163	108	49	8	Yes
2011-2018	31	159	114	48	11	Yes
2012-2019	32	147	120	51	13	Yes
2013-2020	20	58	168	92	25	Yes
Panel B: Variance sorted						
2010-2017	90	157	81	29	6	Yes
2011-2018	103	156	72	27	5	Yes
2012-2019	104	159	74	22	4	Yes
2013-2020	360	2	0	0	1	Yes
Panel C: Relative beta sorted						
2010-2017	35	163	108	49	8	Yes
2011-2018	31	159	114	48	11	Yes
2012-2019	32	147	120	51	13	Yes
2013-2020	20	58	168	92	25	Yes
Panel D: Relative variance sorted						
2010-2017	6	155	132	52	18	Yes
2011-2018	4	161	139	49	10	Yes
2012-2019	4	185	121	42	11	Yes
2013-2020	25	219	91	24	4	Yes

Note: This table summarises the results for the number of stocks providing the highest mean return when sorted on beta, variance, relative beta, and relative variance on a rolling 7-year basis. Panels A, B, C, and D, respectively, show the number of stocks in each quintile when they are sorted by beta, variance, relative beta, and relative variance. The volatility anomaly significantly holds true here since 80%–90% of low-risk and low-volatility stocks provide higher mean returns over a period of 7 years on a rolling basis.

Here, we notice that the highest mean returns are of the stocks in the second and third quintiles when they are beta-sorted, as shown in Figure 8 and the second quadrant of Figure 9. This clearly indicates that low-risk stocks continue to provide higher expected returns for all the 7-year periods taken. The VA claim becomes even stronger when stocks are variance-sorted. The least risky stocks, i.e., the ones belonging to the first and second quintiles, have generated higher returns, especially during the 2013–2020 period. Around 99% of the least risky stocks provided the highest expected returns. Furthermore, the results for relative beta exactly replicate the results obtained for beta-sorted stocks. Thus, we can comfortably infer that the VA is observed for relative beta-sorted stocks for the specific time horizon, as shown in the first quadrant of Figure 9. Finally, in light of the relative variance-sorted stocks, the highest mean returns are obtained for stocks falling under the second and third quintiles. Graphically, this is visible in the fourth quadrant of Figure 9. Therefore, the low-risk stocks fare better than the high-risk ones in terms of average returns for all the 7-year time frames.

Similarly, we cross-verify and exhibit our empirical results for a period of 5 years on a rolling basis, from July 2010 to June 2020. In this case, there are six different subsample data sets generated, i.e., 2010–2015, 2011–2016, 2012–2017, 2013–2018, 2014–2019, and 2015–2020. Figure 10 shows the segregation of stocks yielding the highest mean returns into quintiles when they are sorted on beta values, whereas Figure 11 displays the return pattern of stocks for all values of beta, variance, relative beta, and relative variance for the 5-year period of 2015–2020. The outcomes for the subperiod of 5 years on a rolling basis almost replicate the outcomes obtained for the subperiod of 7 years. Our empirical evidence ascertains the veracity of the VA for a period of 5 years. The data is expressed numerically in Table 2 and graphically in Figures 10 and 11.

Hereafter, we cross-check our results for 3 years on a rolling basis from July 2010 to June 2020, as shown in Table 3. We get eight different subsample data sets, i.e., 2010–2013, 2011–2014, 2012–2015, 2013–2016, 2014–2017, 2015–2018, 2016–2019, and 2017–2020. Figure 12 exhibits the distribution of stocks yielding the highest mean returns into quintiles when they are sorted on beta. Figure 13 depicts the mean return pattern of stocks for all values of beta,

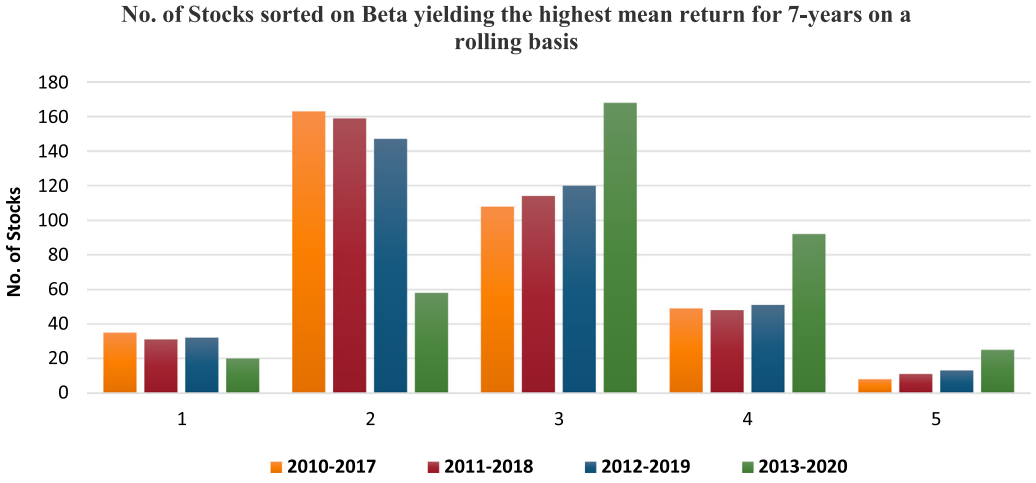


Figure 8 Beta-sorted stocks yielding the highest mean return for 7 years on a rolling basis. Evidently, the VA holds true for all time frames taken, since higher returns are obtained from stocks belonging to the second and third quintiles as compared to the high beta ones in the fourth and fifth quintiles.

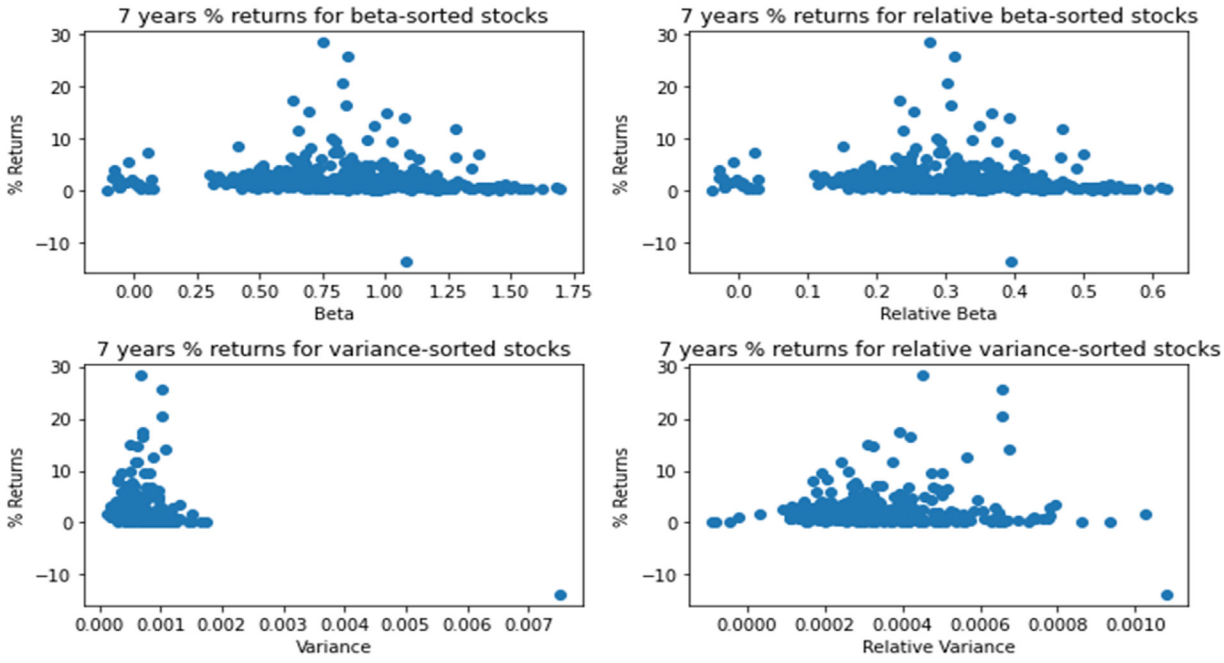


Figure 9 The pattern of returns (in 100s) for all values of risk measures for the 7-year period, 2013-2020. This figure depicts the pattern of mean returns for all values of beta, variance, relative beta, and relative variance for the 7-year period, 2013-2020. The first, second, third, and fourth quadrants represent the return pattern for beta, relative beta, variance, and relative variance sorted stocks, respectively. Clearly, we find that the VA exists for this subsample period of 7 years.

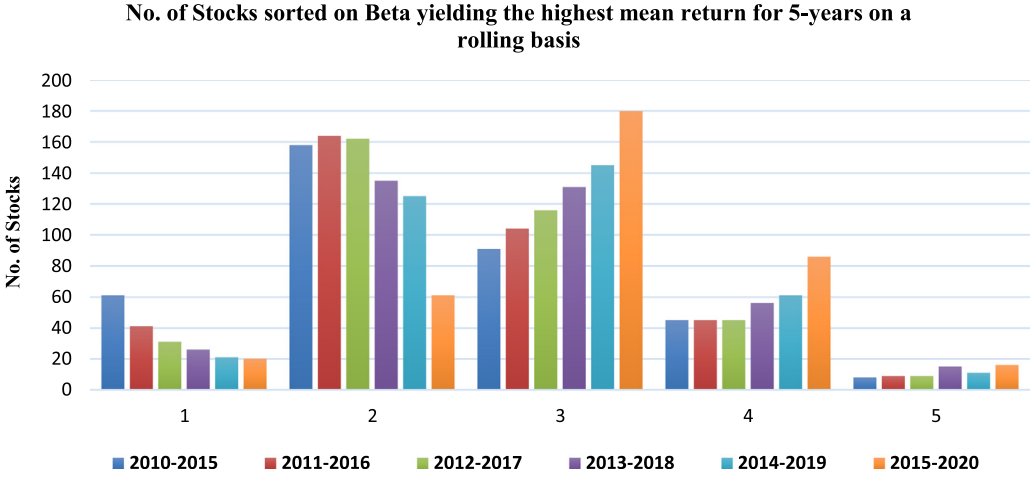


Figure 10 Beta-sorted stocks yielding the highest mean return for 5 years on a rolling basis. This figure exhibits the number of stocks placed in quintiles when they are sorted according to beta for 5 years on a rolling basis from July 2010 to June 2020. There are six different subsample data sets generated, i.e., 2010-2015, 2011-2016, 2012-2017, 2013-2018, 2014-2019, and 2015-2020. Unquestionably, we find that the VA holds true for all time horizons considered since higher and promising returns are obtained from stocks belonging to the second and third quintiles as compared to the high beta ones in the fourth and fifth quintiles.

variance, relative beta, and relative variance for the 3-year period of 2017-2020. As before, we find that the highest returns are generated for stocks belonging to the second and third quintiles, but a noticeable aspect is that higher returns for relative variance-sorted stocks slowly move towards the fourth quintile (especially during the 2013-2016 period). This leads us to question whether VA assuredly existed for a period of 3 years. Although too early, it is reasonable to assume at this point that this anomaly might not always hold true for shorter time horizons. For other risk measures, our outcomes almost replicate the results obtained for the cases of 7 and 5 years. The second and third quintile stocks consistently show higher and promising returns for the majority of the risk measures. The results are depicted numerically in Table 3 and graphically in Figures 12 and 13.

Lastly, we aim to check whether the VA existed for a period of 6 months from January 2020 to June 2020, when the world was hit by the coronavirus pandemic. Table 4 shows the results for the same. Figure 14 graphically displays the return pattern of stocks for all values of beta, variance, relative beta, and relative variance for the 6-month COVID phase from January to June 2020. In light of this rare event,

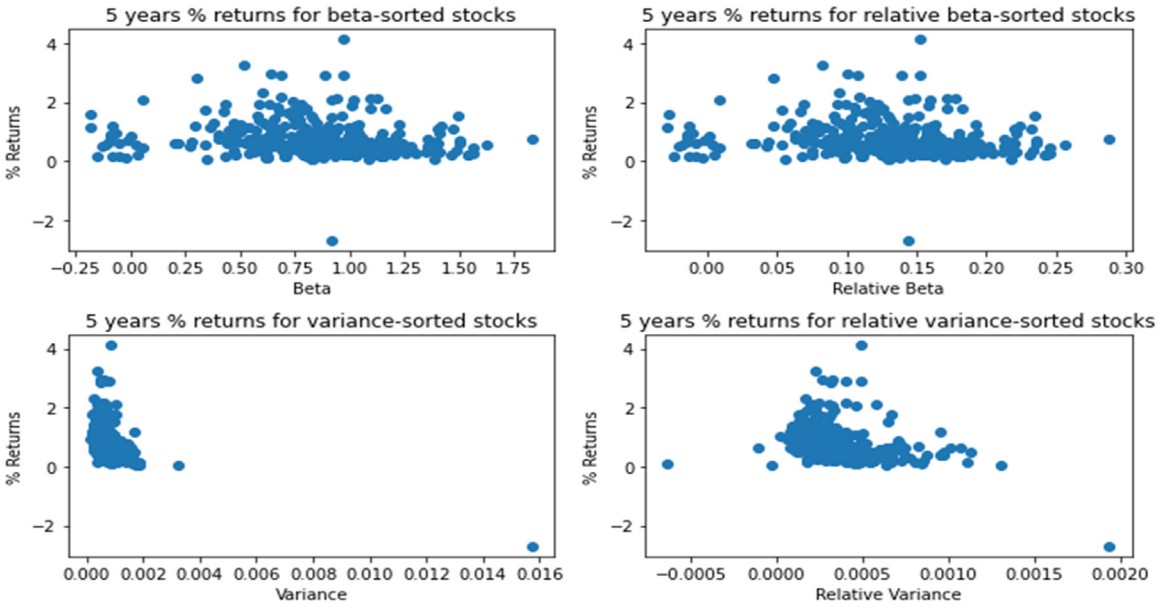


Figure 11 The pattern of returns (in 100s) for all values of risk measures for the 5-year period, 2015–2020. This figure exhibits the pattern of mean returns for all values of beta, variance, relative beta, and relative variance for the 5-year period, 2015–2020. The first, second, third and fourth quadrants represent the return pattern for beta, relative beta, variance, and relative variance sorted stocks, respectively. Undeniably, the VA is present for this subsample period of 5 years.

we find that the highest expected returns have been generated only by stocks belonging to the third quintile when they are sorted according to beta, relative beta, and relative variance. This points to the fact that stocks in the first and second quintiles did not yield high returns in this ultra-short period, as illustrated in Table 4 and Figure 14. This should not be surprising as we observed earlier that VA might not hold true for shorter time frames. Only when stocks are variance-sorted, we find this anomaly to prevail. This motivates us to find the reason underlying the two kinds of outcomes. One of the reasons could be that low-risk stocks can truly reward investors when they are patient since volatility is smoothed and returns are compounded over longer time periods. A factual reason is that in March 2020, the Indian stock market fell by approximately 46% and then recovered significantly by June. Due to such fluctuations, some low-risk stocks generated high returns in the bullish period, while some low-risk stocks fell drastically in value and even went out of business. A few other reasons include the reduction in the policy repo rate by 40 basis points by the Reserve Bank of India and the nationwide lockdown for almost 2 months, which significantly impacted production and increased unemployment.

We also try to explore more generic reasons for the non-existence of the VA in the short run. The average investor in the global equity market probably has a horizon of 3–5 years or even longer (Blitz et al., 2020). The differential treatment of short-term and long-term capital gains tax could be an underlying reason. Short-term capital gains are generally very high in almost all countries when compared to zero or a very low long-term capital gains tax. When investing for the long term, investors would generally prefer low-risk, stable, and fundamentally strong stocks. It is observed that compounding effects cause the long-term return spread between low-risk and high-risk stocks to widen by around 3% annually (Baker et al., 2011). Furthermore, Haritha and Rishad (2020) pointed out that positive sentiment due to irrationality could lead to higher speculative activities and thereby overvaluation of stocks. This sentiment increases short-term volatility but reduces it in the long run due to a bullish outlook on overvalued stocks. All these reasons make a perfect case for the VA to be more of a long-run phenomenon than a short-run one.

Conclusion

This paper revisits the influence of risk and volatility on average stock returns over time. The fact that low-risk stocks generate higher expected returns is a remarkable anomaly in the field of finance. This anomaly has proven to be persistent globally for more than two decades. It contradicts the core concept of traditional finance that higher risks are generally associated with higher rewards. In this study, we have used various established risk measures to present a novel perspective on how the VA exists in the Indian stock market. After accounting for various risk measures and time horizons, we intend to show that anomalous empirical patterns do not necessarily fall in line with traditional asset pricing puzzles.

We observe that the traditional notion of 'higher risks generally yield higher returns' is not always true. This has been proven in equity markets around the world, such as in the United States and China. We attempt to show the existence of low VA, i.e., whether low-risk and low-volatility stocks provide higher expected returns over a stipulated time frame, in the Indian stock market using a cross-section of companies from the NIFTY500 market index.

We first estimated the daily equity returns from the daily stock prices over a period of 10 years, from July 2010 to June 2020. Thereafter, we subject these returns to the

Table 2 Number of stocks yielding the highest mean return sorted on various risk measures (5 years).

	Highest mean return					
	Low				High	
Quintiles 5-year Rolling	1	2	3	4	5	Low > High
Panel A: Beta sorted						
2010-2015	61	158	91	45	8	Yes
2011-2016	41	164	104	45	9	Yes
2012-2017	31	162	116	45	9	Yes
2013-2018	26	135	131	56	15	Yes
2014-2019	21	125	145	61	11	Yes
2015-2020	20	61	180	86	16	Yes
Panel B: Variance sorted						
2010-2015	93	155	79	27	9	Yes
2011-2016	122	146	69	23	3	Yes
2012-2017	139	150	61	12	1	Yes
2013-2018	142	153	55	12	1	Yes
2014-2019	156	153	43	10	1	Yes
2015-2020	361	1	0	0	1	Yes
Panel C: Relative beta sorted						
2010-2015	61	158	91	45	8	Yes
2011-2016	41	164	104	45	9	Yes
2012-2017	31	162	116	45	9	Yes
2013-2018	26	135	131	56	15	Yes
2014-2019	21	125	145	61	11	Yes
2015-2020	20	61	180	86	16	Yes
Panel D: Relative variance sorted						
2010-2015	1	86	196	67	13	Yes
2011-2016	2	55	213	79	14	Yes
2012-2017	2	61	208	76	16	Yes
2013-2018	2	86	203	59	13	Yes
2014-2019	4	204	119	29	7	Yes
2015-2020	3	191	142	23	4	Yes

Note: This table summarises the results for the number of stocks yielding the highest mean return when sorted on beta, variance, relative beta, and relative variance on a rolling 5-year basis. Panels A, B, C, and D, respectively, show the number of stocks in each quintile when they are sorted by beta, variance, relative beta, and relative variance. The volatility anomaly is prominent here since 70–95% of low-risk and low-volatility stocks provide higher mean returns over a period of 5 years on a rolling basis.

Table 3 Number of stocks giving the highest mean return sorted on various risk measures (3 years).

	Highest mean return					
	Low				High	
Quintiles 3-year Rolling	1	2	3	4	5	Low > High
Panel A: Beta sorted						
2010-2013	48	149	107	45	14	Yes
2011-2014	83	148	76	43	13	Yes
2012-2015	75	162	82	36	8	Yes
2013-2016	29	163	110	51	10	Yes
2014-2017	23	122	142	67	9	Yes
2015-2018	22	99	157	71	14	Yes
2016-2019	22	114	151	62	14	Yes
2017-2020	22	64	183	82	12	Yes
Panel B: Variance sorted						
2010-2013	151	140	53	13	6	Yes
2011-2014	146	144	54	15	4	Yes
2012-2015	133	150	60	17	3	Yes
2013-2016	178	140	35	9	1	Yes
2014-2017	216	122	22	2	1	Yes
2015-2018	231	110	20	1	1	Yes
2016-2019	129	156	54	14	10	Yes
2017-2020	362	0	0	0	1	Yes
Panel C: Relative beta sorted						
2010-2013	48	149	107	45	14	Yes
2011-2014	83	148	76	43	13	Yes
2012-2015	75	162	82	36	8	Yes
2013-2016	29	163	110	51	10	Yes
2014-2017	23	122	142	67	9	Yes
2015-2018	22	99	157	71	14	Yes
2016-2019	22	114	151	62	14	Yes
2017-2020	22	64	183	82	12	Yes
Panel D: Relative variance sorted						
2010-2013	123	163	56	14	7	Yes
2011-2014	1	0	195	145	22	Yes
2012-2015	1	0	229	124	9	Yes
2013-2016	2	1	147	182	31	No
2014-2017	1	3	263	86	10	Yes
2015-2018	1	6	280	67	9	Yes
2016-2019	6	255	80	17	5	Yes
2017-2020	1	258	94	9	1	Yes

Note: This table summarises the results for the number of stocks providing the highest mean return when sorted on beta, variance, relative beta, and relative variance on a rolling 3-year basis. Panels A, B, C, and D, respectively, show the number of stocks in each quintile when they are sorted by beta, variance, relative beta, and relative variance. The volatility anomaly clearly holds true here since 65%–95% of low-risk and low-volatility stocks provide a higher mean return over a period of 3 years on a rolling basis.

treatment of various established risk standards like beta, variance, relative beta, relative variance, *VaR*, downside semi-variance, and downside beta. We divided the relevant data into quintiles (1 is the bottom quintile and 5 is the top) based on the values of these risk measures. For the 10-year period, our results confirm the existence of the VA. Our results show that low-risk stocks (i.e., with low beta, variance, relative beta, and relative variance) yield higher expected returns than high-risk stocks. To cross-verify these findings for smaller time intervals, we have repeated this procedure for a period of 7 years, 5 years, and 3 years on a rolling basis. We observe that this anomaly undoubtedly exists for cases of 7 and 5 years on a rolling basis. However, for the short period of 3 years and the ultra-short period of 6 months (January to June 2020) during the COVID lockdown period, this anomaly could not be confirmed. Hence, we can observe that the VA is a medium- to long-term phenomenon but not necessarily a short-term one. A satisfactory explanation is that the market typically requires 3 years or more for risk adjustment and volatility smoothing to accrue higher returns.

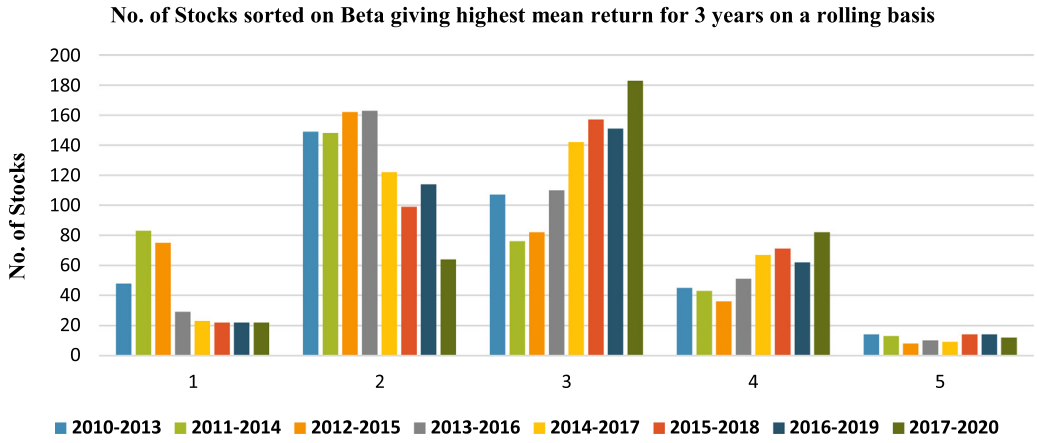


Figure 12 Beta-sorted stocks giving the highest mean return for 3 years on a rolling basis. This figure exhibits the number of stocks in each of the five quintiles when they are sorted on beta for 3 years on a rolling basis from July 2010 to June 2020. There are eight different subsample data sets, i.e., 2010-2013, 2011-2014, 2012-2015, 2013-2016, 2014-2017, 2015-2018, 2016-2019 and 2017-2020. Assuredly, we find that the VA holds true for all time horizons considered since higher mean returns are achieved from stocks belonging to low beta quintiles (second and third) as compared to the high beta ones (fourth and fifth quintiles).

Emerging markets are generally known to be highly volatile in nature. Therefore, foreign investors are perpetually concerned with risk-adjusted returns, especially when they are investing in emerging markets like India. This study shows that despite the overall high volatility in emerging markets like India, there is an opportunity to generate high risk-adjusted returns due to the existence of the VA. The bonus from the low-risk-high-return strategies is attributable to the compounding effect – the lower volatility pressure on investment returns enhances the performance of less volatile stocks. The real benefit of possessing low-volatility stocks accrues over longer periods of time, as the well-known power of compounding suggests (Ang et al., 2006). Furthermore, our results can be useful for investors who wish to invest in Asian countries. As most of the Asian countries are emerging economies (except a few), the existence of the VA may not be limited to only India. We can expect similar patterns in other Asian countries. Foreign (or even domestic) investors can use this anomaly to formulate low-risk-high-return strategies for investing in Asian financial markets.

This paper holds considerable potential for further development. For instance, this anomaly can be verified for equity markets in major developed, developing, and emerging economies all over the world. This study can be extended by using other risk proxies such as expected shortfall, jump risk, etc. Apart from risk and volatility measures, other

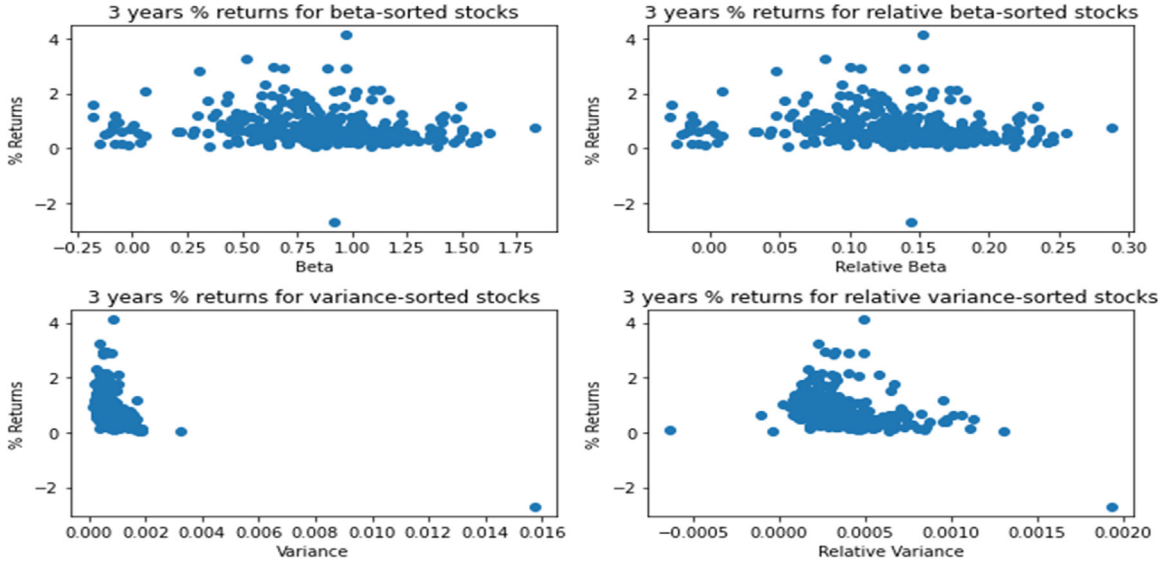


Figure 13 The pattern of returns (in 100s) for all values of risk measures for the 3-year period 2017-2020. This figure presents the return pattern for all values of beta, variance, relative beta, and relative variance for the 3-year sample period, 2017-2020. The first, second, third and fourth quadrants represent the return pattern for beta, relative beta, variance, and relative variance sorted stocks, respectively. Except for relative variance, all other risk measures substantiate the existence of the VA for this subsample period of 3 years. For the period 2013-2016, the stocks in the fourth and fifth quintiles exhibit higher returns than the low relative variance-sorted quintiles.

Table 4 Number of stocks giving the highest mean return sorted on various risk measures (6 months).

	Highest mean return					
	Low				High	
Quintiles 6-month COVID time	1	2	3	4	5	Low > High
Panel A: Beta sorted						
Jan–June 2020	19	43	177	106	18	Yes
Panel B: Variance sorted						
Jan–June 2020	283	76	3	0	1	Yes
Panel C: Relative Beta sorted						
Jan–June 2020	19	43	177	106	18	Yes
Panel D: Relative variance sorted						
Jan–June 2020	1	98	234	25	5	Yes

Note: This table summarises the results for a number of stocks providing the highest mean return when sorted on beta, variance, relative beta and relative variance for a period of 6 months when the world witnessed the coronavirus pandemic. Panels A, B, C, and D, respectively, show the number of stocks in each quintile when they are sorted by beta, variance, relative beta, and relative variance. The volatility anomaly does not exist for this subsample period of 6 months since the highest returns are found only in stocks falling under the 3rd quintile, not the 1st or even the 2nd quintile. Variance-sorted show the opposite picture.

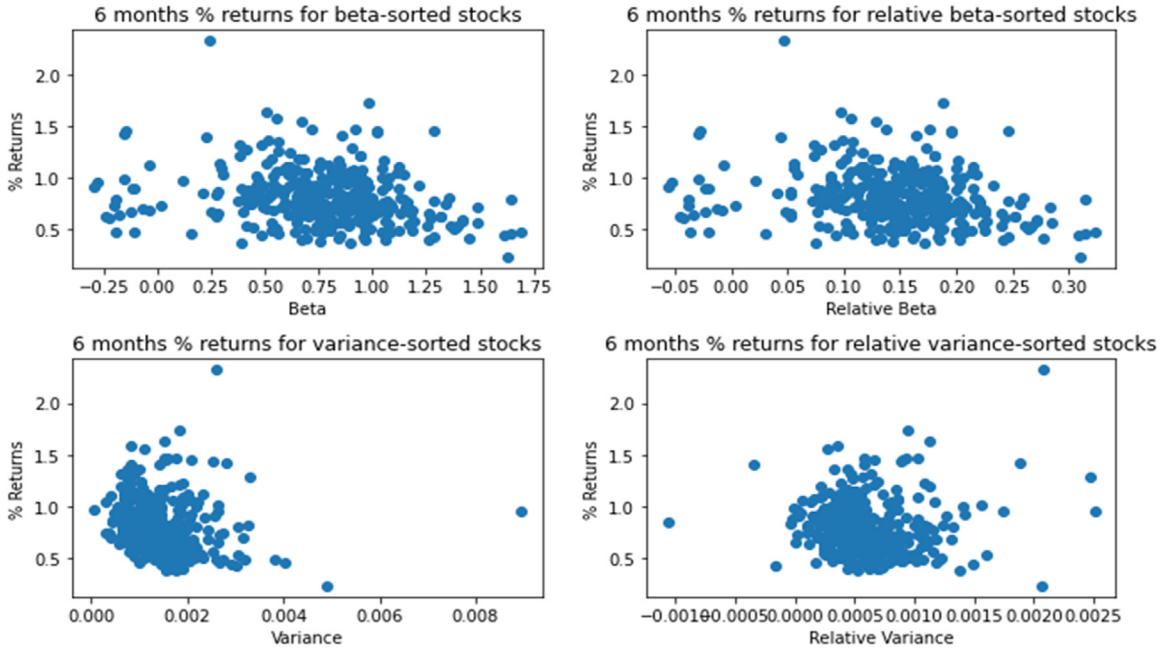


Figure 14 The pattern of returns (in 100s) for all values of risk measures for 6 months from January 2020 to June 2020. The following graph presents the return pattern for all values of beta, variance, relative beta, and relative variance for the 6-month subsample period, January–June 2020. The first, second, third, and fourth quadrants represent the return pattern for beta, relative beta, variance, and relative variance sorted stocks, respectively. Except for variance, all other risk measures verify the non-existence of the VA for this ultra-short period. In the case of variance-sorted stocks, the results are contradictory.

factors like skewness asymmetry of stock returns, betting-against-beta strategy, cash flow variability, size, and quality of the firm, and so on can also be used to further explore and gain clarity on the VA phenomenon. These techniques are inspired by the works of Atilgan, Demirtas, and Gunaydin (2020), Dutt and Humphery-Jenner (2013), Frazzini and Pedersen (2014), and Walkshäusl (2013).

Data availability statement

The data that supports the findings of this study are openly available at https://in.finance.yahoo.com/. The simulation code and data that support the findings of this study are available from the corresponding author upon reasonable request.

Code availability

The code used in this study is available from the corresponding author upon reasonable request.

Declaration of Competing Interest

Authors have no conflict of interest to declare.

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References

Ang, A., Chen, J., & Xing, Y. (2006). Downside risk. *The Review of Financial studies* 19 (4), 1191–1239.

Ang, A., Hodrick, R.J., Xing, Y., & Zhang, X. (2006). The cross-section of volatility and expected returns. *The Journal of Finance* 61 (1), 259–299.

Ang, A., Hodrick, R.J., Xing, Y., & Zhang, X. (2006). The cross—section of volatility and expected returns. *The journal of finance* 61 (1), 259–299.

Ang, A., Hodrick, R.J., Xing, Y., & Zhang, X. (2009). High idiosyncratic volatility and low returns: International and further US evidence. *Journal of Financial Economics* 91 (1), 1–23.

Atilgan, Y., Bali, T.G., Demirtas, K.O., & Gunaydin, A.D. (2018). Downside beta and equity returns around the world. *The Journal of Portfolio Management* 44 (7), 39–54.

Atilgan, Y., Demirtas, K.O., & Gunaydin, A.D. (2020). Downside beta and the cross section of equity returns: A decade later. *European Financial Management* 26 (2), 316–347.

Baker, M., Bradley, B., & Wurgler, J. (2011). Benchmarks as limits to arbitrage: Understanding the low-volatility anomaly. *Financial Analysts Journal* 67 (1), 40–54.

Baker, N. L., & Haugen, R. A. (2012). Low risk stocks outperform within all observable markets of the world. Available at: SSRN 2055431.

Barro, R.J. (2006). Rare disasters and asset markets in the twentieth century. *The Quarterly Journal of Economics* 121 (3), 823–866.

Blitz, D.C., & van Vliet, P. (2007). The volatility effect. *The Journal of Portfolio Management* 34 (1), 102–113.

Black, F. (1972). Capital market equilibrium with restricted borrowing. *The Journal of business* 45 (3), 444–455.

Blitz, D., Huisman, R., Swinkels, L., & van Vliet, P. (2020). Media attention and the volatility effect. *Finance Research Letters* 36, 101317.

Boasson, V., Boasson, E., & Zhou, Z. (2011). Portfolio optimization in a mean-semivariance framework. *Investment management and Financial innovations* 8 (3), 58–68.

Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *The review of economics and statistics* 69, 542–547.

Brennan, M.J. (1971). Capital market equilibrium with divergent borrowing and lending rates. *Journal of Financial and Quantitative Analysis* 6 (5), 1197–1205.

Campbell, J.Y., & Viceira, L.M. (1999). Consumption and portfolio decisions when expected returns are time varying. *The Quarterly Journal of Economics* 114 (2), 433–495.

Collver, C. D., De Jong, J., & Palkar, D. (2013). Risk and the volatility anomaly: A global analysis. Available at: SSRN 2236693.

Clarke, R., De Silva, H., & Thorley, S. (2006). Minimum-variance portfolios in the US equity market. *Journal of Portfolio Management* 33 (1), 10.

Core, J.E., Guay, W.R., & Rusticus, T.O. (2006). Does weak governance cause weak stock returns? An examination of firm operating performance and investors' expectations. *The Journal of Finance* 61 (2), 655–687.

Danielsson, J., & De Vries, C.G. (2000). Value-at-risk and extreme returns. *Annales d'Economie et de Statistique* 60, 239–270.

De Jong, J., & Palkar, D.D. (2016). Risk and the Volatility anomaly. *The Journal of Investing* 25 (3), 17–28.

Dicle, M.F., Beyhan, A., & Yao, L.J. (2010). Market efficiency and international diversification: Evidence from India. *International Review of Economics & Finance* 19 (2), 313–339.

Dutt, T., & Humphery-Jenner, M. (2013). Stock return volatility, operating performance and stock returns: International evidence on drivers of the 'low volatility' anomaly. *Journal of Banking & Finance* 37 (3), 999–1017.

Estrada, J. (2002). Systematic risk in emerging markets: the D-CAPM. *Emerging Markets Review* 3 (4), 365–379.

Estrada, J. (2007). Mean-semivariance behavior: Downside risk and capital asset pricing. *International Review of Economics & Finance* 16 (2), 169–185.

Fama, E.F., & French, K.R. (1992). The cross-section of expected stock returns. *The Journal of Finance* 47 (2), 427–465.

Fama, E.F., & MacBeth, J.D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of political economy* 81 (3), 607–636.

Fong, W.M. (2018). "Safe" stocks. *Journal of Asset Management* 19 (2), 93–98.

Frazzini, A., & Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics* 111 (1), 1–25.

Gabaix, X. (2008). Variable rare disasters: A tractable theory of ten puzzles in macro-finance. *American Economic Review* 98 (2), 64–67.

Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *The Quarterly Journal of economics* 127 (2), 645–700.

Grootveld, H., & Hallerbach, W. (1999). Variance vs downside risk: Is there really that much difference? *European Journal of operational research* 114 (2), 304–319.

Haritha, P.H., & Rishad, A. (2020). An empirical examination of investor sentiment and stock market volatility: evidence from India. *Financial Innovation* 6 (1), 1–15.

Haugen, R.A., & Heins, A.J. (1975). Risk and the rate of return on financial assets: Some old wine in new bottles. *Journal of Financial and Quantitative Analysis* 10 (5), 775–784.

Heston, S.L., Rouwenhorst, K.G., & Wessels, R.E. (1999). The role of beta and size in the cross-section of European stock returns. *European Financial Management* 5 (1), 9–27.

Jensen, M.C., Black, F., & Scholes, M.S. (1972). The capital asset pricing model: Some empirical tests. *Studies in the Theory of Capital Markets*. Praeger, New York.

Koch, S. (2010). Low risk and high returns: Evidence from the German stock market. Available at: SSRN 1632425.

Kumar, K., Mamidi, V., & Marisetty, V. (2011). Global markets exposure and price efficiency: An empirical analysis of order flow dynamics of NYSE-listed Indian firms. *Journal of International Financial Markets, Institutions and Money* 21 (5), 686–706.

Lakonishok, J., & Shapiro, A.C. (1984). Stock returns, beta, variance and size: An empirical analysis. *Financial Analysts Journal* 40 (4), 36–41.

Leibowitz, M.L. (1986). Total portfolio duration: a new perspective on asset allocation. *Financial Analysts Journal* 42 (5), 18–29.

Lintner, J. (1965). Security prices, risk, and maximal gains from diversification. *The journal of finance* 20 (4), 587–615.

Markowitz, H. (1959). Portfolio selection, efficient diversification of investments. John Wiley & Sons.

Narayan, P.K., & Ahmed, H.A. (2014). Importance of skewness in decision making: Evidence from the Indian stock exchange. *Global Finance Journal* 25 (3), 260–269.

Narayan, P.K., Narayan, S., & Prabheesh, K.P. (2014). Stock returns, mutual fund flows and spillover shocks. *Pacific-Basin Finance Journal* 29, 146–162.

Narayan, P.K., Ahmed, H.A., & Narayan, S. (2015). Do momentum-based trading strategies work in the commodity futures markets? *Journal of Futures Markets* 35 (9), 868–891.

Post, T., Van Vliet, P., & Lansdorp, S. (2012). Sorting out downside beta. Available at: SSRN 1980614.

Reinganum, M.R. (1981). Misspecification of capital asset pricing: Empirical anomalies based on earnings' yields and market values. *Journal of Financial Economics* 9 (1), 19–46.

Rietz, T.A. (1988). The equity risk premium a solution. *Journal of Monetary Economics* 22 (1), 117–131.

Saengchote, K. (2017). The low-risk anomaly: evidence from the Thai stock market. *Asian Academy of Management Journal of Accounting & Finance* 13 (1), 143–158.

Sharpe, W.F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The journal of finance* 19 (3), 425–442.

Shefrin, H. (2001). Do investors expect higher returns from safer stocks than from riskier stocks? *The Journal of Psychology and Financial Markets* 2 (4), 176–181.

Shleifer, A., & Vishny, R.W. (1997). The limits of arbitrage. *The Journal of Finance* 52 (1), 35–55.

Stambaugh, R.F., Yu, J., & Yuan, Y. (2015). Arbitrage asymmetry and the idiosyncratic volatility puzzle. *The Journal of Finance* 70 (5), 1903–1948.

Vuolteenaho, T. (2002). What drives firm-level stock returns? *The Journal of Finance* 57 (1), 233–264.

Walkshäusl, C. (2013). The high returns to low volatility stocks are actually a premium on high quality firms. *Review of Financial Economics* 22 (4), 180–186.